GENERALIZED SAMPLING SYSTEM

Version 1.0

METHODOLOGY GUIDE

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Business Survey Methods Division

G-SAM

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Foreword

Purpose

This document is the Methodology Guide for version 1.0 of the Generalized Sampling System (G-SAM). It gives a detailed description of the methodology used for each G-SAM module: stratification, allocation, sampling and coordination.

G-SAM

G-SAM is a SAS® based application. The G-SAM software consists of a set of SAS macros. The source code is written using the SAS programming language. For a detailed description of how to use the G-SAM functionalities please consult the G-SAM User Guide.

Credits

Documentation

Writing

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CONTENTS

G-SAM Modules

1. Stratification

The stratification module built for version G-SAM version 1.0 implements two simple stratification methods: geometric (GunningHorgan2004) and cumulative square root f (DaleniusHodges1959). It also implements the exclusion threshold method (MarandaRoyce 1998) –also known as the Royce-Maranda method– used at STC to determine the take none portion in business surveys. Since there are no optimal stratification procedures in this version, the methodology is rather straightforward.

1.1. Cumulative Square Root f Method

This method determines stratum boundaries $b_0, b_1, \ldots, b_H$ so that every stratum has about the same cumulative square root frequency. Assuming the values are approximately uniformly distributed within each stratum, this choice aims at minimizing the theoretical sampling variance for a density f when sampling independently in each stratum.

The basic implementation goes as follows. We begin by defining groups called *cells*, which correspond to the distinct values taken by some classification variables available on the population file. In each cell the values of the variable of interest are arranged in ascending order $y_{(1)}, y_{(2)}, \ldots, y_{(N_c)}$ and then grouped into G equally spaced intervals. To produce meaningful results, the parameter G should be chosen much larger–but not too large– than H, the desired number of substrata in each cell. Next, we determine the frequency f_g of each interval and then compute $\sqrt{f_g}$. For each interval we then obtain the cumulative square root sum $S_g = \sum_1^g f_i^{1/2}$. For each substratum h<H, define the upper boundary b_h to be the largest $y_{(i)}$ value corresponding to the cumulative square root sum S_g that is closest to hS_G/H . Once the boundaries $b_1, \ldots, b_{H-1}$ have been obtained, take $b_0 = y_{(1)}$, $b_H = y_{(N)}$ and form the sub strata as $[b_0, b_1), [b_1, b_2) \ldots, [b_{H-1}, b_H]$.
These steps are repeated until all cells have been partitioned. The final strata are the different combinations cell-substratum.

Sometimes the user may also want to exclude cell outliers from this process and put them in their own strata. For that purpose, we provide additional parameters a and b to specify percentiles within the cell. The respective percentile values y_c^{inf} , y_c^{sup} define boundaries for the additional sub strata in the following way: $[b_0, b_1) = \left[y_{(1)}, y_c^{inf}\right)$ and/or $[b_{H-1}, b_H] = \left[y_c^{sup}, y_{(N)}\right]$.

The basic recipe is then applied to all cell values in $[\,y_c^{inf}, y_c^{sup}]$ adjusting the number of sub strata so that in the end we have H sub strata for the entire cell. For convenience, we also allow the users to specify integer values for the parameters a and b. This represents only a minor variation of the above scheme. When $a = k$, and/or $b = m$, we exclude the k^ith smallest cell values and/or the m^th largest and put them in their own strata.
Despite the shortcuts used in the notation, the parameters H, G, a and b are cell specific. Users can provide a mixture of global and cell specific parameters through the appropriate input files.

1.2. Geometric Method

This method determines initial stratum boundaries $b_0, b_1, \dots, b_H$ that correspond to the first terms of a geometric series for which the extreme terms coincide with the minimum and the maximum values of the variable of interest in the cell.

The basic implementation goes as follows. As for the cumulative square root method we begin by defining groups called *cells* which correspond to the distinct values taken by some classification variables available on the population file. In each cell the values of the variable of interest are arranged in ascending order $y_{(1)}, y_{(2)}, \dots, y_{(N_c)}$. We then find the first positive value $y_c^+ = min\{y_{(i)}: y_{(i)} > 0\}$ and compute the ratio $r_c = \left[y_{(N)}/y_c^+\right]^{1/H}$. The initial boundaries correspond to the successive values $x_h = y_c^+ r_c^h, h = 1, \dots, H-1$. These values could be used to define the stratum boundaries, but in this case some strata may end up being empty. Instead, in each substratum $h < H$, we take the upper boundary b_h as the $y_{(i)}$ value closest to x_h. That said, for large problems the former approach is considerably faster.
Once the above upper boundaries have been determined, we define $b_0 = y_{(1)}$, $b_H = y_{(N)}$ and form the sub strata as $[b_0, b_1), [b_1, b_2), \dots, [b_{H-1}, b_H]$. These steps are repeated until all cells have been partitioned. The final strata are the different combinations cell-substratum.

As before, users may want to exclude cell outliers from this process and put them in their own strata. For that purpose, we provide additional parameters a and b to specify percentiles within the cell. The respective percentile values y_c^{inf} , y_c^{sup} define boundaries for the additional sub strata in the following way: $[b_0, b_1) = [y_{(1)}, y_c^{inf})$ and/or $[b_{H-1}, b_H] = [y_c^{sup}, y_{(N)}]$. The basic recipe is then applied to all cell values in $[\,y_c^{inf}, y_c^{sup}]$ adjusting the number of sub strata so that in the end we have H sub strata for the entire cell. For convenience we also want to allow the user to specify integer values for the parameters a and b. When $a = k$, and/or $b = m$, we then have to exclude the k^ith smallest cell values and/or the m^th largest and put them in their own strata.

Another option users have asked for is the possibility to exclude certain units from the computation of the geometric sequence (x_h). Those units would be flagged on the population file. All the steps remain the same as for the basic method, except that the computation of y_c^+ and r_c exclude the cell units that have been flagged.

As always, despite the shortcuts used in the notation, the parameters H, a and b are cell specific. If the user provides a mixture of global and cell specific parameters (through the appropriate input files), the cell specific values always prevail.

1.3. Take-None Portion

Rather than building an entire partition of the population like in the other methods, the objective here is to determine a portion, called the *take-none portion*, that will be excluded from the sampling process. The units that form the take-none portion are typically small units that represent a negligible fraction of the population total, and for which auxiliary information is available to replace survey data. The resulting exclusion threshold depends on a nominal exclusion level α , nominal threshold levels (λ_i) and a tolerance parameter ε.

As for the previous methods we begin by defining groups called *cells* which correspond to the distinct values taken by some classification variables available on the population file. In each cell the values of the variable of interest are arranged in ascending order $y_{(1)}, y_{(2)}, \dots, y_{(N_c)}$. Then, we determine the largest value q_α such that for the nominal threshold α we do have $\sum_{y_{(i)} \geq q_\alpha} y_{(i)} \geq (1-\alpha)T_c$, where T_c represents the cell total. This corresponds to the weighted percentile of level α for the cell when we use the weights $\omega_i = y_i/T_c$. Similarly, we determine the weighted percentile $q_{\alpha+\varepsilon}$ where ε is the tolerance level. If $\lambda_i > q_\alpha$ for all λ_i , we define the exclusion threshold as $\lambda_c = min\{ \lambda_i \}$. Otherwise, we find the largest λ_i such that $\lambda_i \leq q_\alpha$. If we also have $\lambda_{i+1} \leq q_{\alpha+\varepsilon}$, then we take $\lambda_c = \lambda_{i+1}$, else we take $\lambda_c = \lambda_i$. The take-none units in the cell are those that satisfy the condition $y_i < \lambda_c$. The effective exclusion fraction achieved with λ_c is $f_c = \sum_{y_{(i)} < \lambda_c} y_{(i)}/T_c$. These steps are repeated until all cells have been covered, and the take-none portion is formed by the take-none units in all the cells.

2. Allocation

The allocation module that was developed for G-SAM uses a constrained optimization approach in order to obtain allocations that have optimal properties under some ideal sampling strategies. Even though these ideal setups are seldom encountered in the survey practice, the resulting allocations are still quite useful. In what follows we will frequently borrow from standard optimization theory vocabulary to describe the various features of our allocation problems. However, we do not assume the user to be familiar with these notions and the meaning of expressions such as objective function, constraints, resources, feasible solution, etc., will become clear from the context.

Sampling variances and CV's are another key ingredient for building the optimisation models used for allocating the sample. For reasons that will become clear later on, we will not use standard variance formulas based on linearization techniques and so we first need to spend some time deriving ones which are more suitable for our needs. Once this is done we will be able to write down our allocation problems and describe the strategies that were implemented in order to solve them. Although not without interest, these details are not essential for the description of the optimal allocation methods and the reader that desires so can go directly to section 2.5.

2.1. Sampling designs, Estimators and Variances

Two basic sample selection schemes were considered for optimal allocation: 2-phase stratified simple random sampling (SRS) and 2-phase stratified Bernoulli sampling (BER). These sampling plans are characterized by the fact that sampling is made on (up to) two occasions and the units are always of the same type—for users familiar with StatMx terminology those designs belong to the one-stage 2-phase element type (E2). They present the useful feature that when the first phase is a census, the corresponding sampling variances reduce to their basic 1-phase counterparts, so our allocation will be able to handle these important cases as well.
Further, provided the total estimator is adjusted for the non-response this approach allows us to incorporate a simple non-response model into our equations. Before going into the details, let us first introduce some of the notation that will be used in our account.

Given a population U of units k with characteristics y_k, we are interested in point estimates for the domain total $t_d = \sum_D y_k$. To avoid technical pitfalls we will always suppose the domain D to be a fixed subset of U –not random. The estimates we consider are of the form $\hat{t}_d = \sum_S \frac{y_{dk}}{\alpha_k}$, where S is the set of units in the sample $y_{dk} = y_k \cdot \mathrm{I}(k \in D)$ and the coefficients α_k are chosen in accordance to our sampling plan– we will leave them unspecified for the moment. Various types of indicators will also be used, often they will be denoted $\mathrm{I_k}$ but the definition will change depending on the context.

The next lemma is a useful generalization of a well known result about inclusion probabilities for fixed sample size designs (Särndal, Swensson and Wretman (1992), result 2.6.2). The fact that it can be used on arbitrary designs makes it quite handy.

Lemma1. Let $\mathrm{I_1}, \dots, \mathrm{I_N}$ be exchangeable indicator random variables such that $\sum_1^N \mathrm{I_j} = n_s$. Then for all $j \neq k \epsilon \{1, \dots, N\}$ we have $E\left(\mathrm{I_j}|n_s\right) = \frac{n_s}{N}$ and $E\left(\mathrm{I_j I_k}|n_s\right) = \frac{n_s(n_s-1)}{N(N-1)}$.

PROOF. Exchangeable random variables are those for which the joint distribution is symmetrical in its arguments. The exchangeability condition implies that $\mathrm{I_1}, \dots, \mathrm{I_N}$ are conditionally exchangeable given $\sum_1^N \mathrm{I_j}$. The result then follows from basic computations involving standard properties of conditional expectations:

$$n_s = E(\textstyle\sum_1^N \mathrm{I_k}\,|n_s) = NE(\mathrm{I_k}|n_s)\,, \quad n_s^2 = E\left[\left(\sum_1^N \mathrm{I_j}\right)^2 |n_s\right] = E\left(\sum_1^N \mathrm{I_j}\,|n_s\right) + \sum\sum_{j\neq k} E\left(\mathrm{I_j I_k}|n_s\right)$$
$$= n_s + N(N-1)E\left(\mathrm{I_j I_k}|n_s\right) \blacksquare$$

2.2. SRS with MAR non-response

Our first task is to show the steps for computing the sampling variance of the estimator of the total of a domain under an SRS design and a basic missing at random (MAR) non-response model. Of course, at this stage there is no imputation and we deal with non-response only by reweighting the respondents. In this context, our estimator becomes $\hat{t}_d = \sum_{S_r} \frac{y_{dk}}{\alpha_\mathrm{k}}$, where $\alpha_\mathrm{k} = n_r/N$ and n_r is the number of respondents. In fact, this estimator is not the usual Horvitz-Thompson estimator but rather a calibration estimator where we calibrate on the population count. Using the first part of Lemma 1, we obtain the conditional expectation given n_r .

$$E\left[N \sum_{S_r} \frac{y_{dk}}{n_r} |n_r\right] = E\left[N \sum_U \frac{y_{dk}\mathrm{I}_k}{n_r} |n_r\right] = N \sum_U \frac{y_{dk}\mathrm{E}(\mathrm{I}_k|n_r)}{n_r} = \sum_U y_{dk} = N\bar{y}_d,$$

where $\mathrm{I_k}$ is the indicator of the event $\{\mathrm{k} \in S_r\}$. Since the sampling design is SRS and the responses are i.i.d. Bernoulli, the joint distribution of these indicators is clearly symmetrical in its arguments so the lemma applies.

The conditional expectation of $\hat{\mathrm{t}}_\mathrm{d}^2$ given n_r is obtained in a similar way

$$E\left[\frac{\mathrm{N}^2}{n_r^2}\left(\sum_{S_r} \frac{y_{dk}}{n_r}\right)^2 |n_r\right] = E\left[\sum_U \sum_U \frac{\mathrm{N}^2 y_{dj} y_{dk} \mathrm{I}_j \mathrm{I}_k}{n_r^2} |n_r\right]$$
$$= \sum_U \frac{\mathrm{N}^2 y_{dk}^2}{n_r^2} \mathrm{E}(\mathrm{I}_k|n_r) + \sum\sum_{j\neq k} \frac{\mathrm{N}^2 y_{dj} y_{dk} \mathrm{E}(\mathrm{I}_j \mathrm{I}_k|n_r)}{n_r^2}$$
$$= \sum_U \frac{N y_{dk}^2}{n_r} + \sum\sum_{j\neq k} \frac{N y_{dj} y_{dk}}{(N-1)}\left[1 - \frac{1}{n_r}\right]$$
$$= \frac{\mathrm{N}}{n_r}\left[\sum_U y_{dk}^2 - \sum\sum_{j\neq k} \frac{y_{dj} y_{dk}}{(N-1)}\right] + \sum\sum_{j\neq k} \frac{N y_{dj} y_{dk}}{(N-1)}$$

$$= \frac{\mathrm{N}^2}{n_r} S_{yd}^2 + \sum\sum_{j \neq k} \frac{N y_{dj} y_{dk}}{(N-1)}$$

Here it is implicit that $n_r > 0$, but since we also have to handle the case $n_r = 0$, we define all relevant quantities to be equal to zero when that happens:

$$E[\,\hat{t}_d |\, n_r] = N\bar{y}_d \mathrm{I}\{n_r > 0\} \quad \text{and} \quad E\left[\,\hat{\mathrm{t}}_{\mathrm{d}}^2 \mid n_r\right] = \begin{cases} \frac{\mathrm{N}^2}{n_r} S_{yd}^2 + \sum\sum_{j \neq k} \frac{N y_{dj} y_{dk}}{(N-1)}, n_r > 0 \\ 0, \;\; n_r = 0 \end{cases} \qquad (2.1.1)$$

Taking expectations, we obtain

$$E[\,\hat{t}_d] = N\bar{y}_d \mathrm{P}\{n_r > 0\}\;, \quad E\left[\,\hat{\mathrm{t}}_{\mathrm{d}}^2\,\right] = N^2 E\left(n_r^{-1} \mathrm{I}_{\{n_r > 0\}}\right) S_{yd}^2 + \mathrm{P}\{n_r > 0\} \sum\sum_{j \neq k} \frac{N y_{dj} y_{dk}}{(N-1)}$$

We now have all the ingredients to compute the variance

$$V\left[N \sum_{S_r} \frac{y_{dk}}{n_r}\right] = N^2 E\left(n_r^{-1} \mathrm{I}_{\{n_r > 0\}}\right) S_{yd}^2 + \mathrm{P}\{n_r > 0\} \sum\sum_{j \neq k} \frac{N y_{dj} y_{dk}}{(N-1)} - N^2 \bar{y}_d^2 \mathrm{P}\{n_r > 0\}^2$$

$$= N^2 E\left(n_r^{-1} \mathrm{I}_{\{n_r > 0\}}\right) S_{yd}^2 + \mathrm{P}\{n_r > 0\} \left\{\sum\sum_{j \neq k} \frac{N y_{dj} y_{dk}}{(N-1)} - N^2 \bar{y}_d^2\right\} + \mathrm{P}\{n_r > 0\} \mathrm{P}\{n_r = 0\} N^2 \bar{y}_d^2$$

$$= N^2 E\left(n_r^{-1} \mathrm{I}_{\{n_r > 0\}}\right) S_{yd}^2 - \mathrm{P}\{n_r > 0\} N S_{yd}^2 + \mathrm{P}\{n_r > 0\} \mathrm{P}\{n_r = 0\} N^2 \bar{y}_d^2$$

$$= N^2 \left[E\left(n_r^{-1} \mathrm{I}_{\{n_r > 0\}}\right) - \frac{\mathrm{P}\{n_r > 0\}}{N}\right] S_{yd}^2 + \mathrm{P}\{n_r > 0\} \mathrm{P}\{n_r = 0\} N^2 \bar{y}_d^2 \qquad (2.1.2)$$

When we have full response, the above formula yields the correct variance since in that case $P(n_r > 0) = 1$ and the term that contains the expectation is just n^{-1}. This also gives the correct variance for the estimation strategy consisting of Bernoulli sampling combined with the Hajek estimator, as we can make our SRS sample to be the entire population and replace the non response mechanism by the Bernoulli selection.

From (2.1.2) we observe that a nuisance term containing $\bar{y}_d^2$ appears when $\mathrm{P}\{n_r = 0\} > 0$. If in a Bernoulli design (with no non-response) we choose to reject the event $\{n_r = 0\}$ whenever it occurs, it can be seen that the conditional expectations now become

$$E[\,\hat{t}_d |\, n_r] = N\bar{y}_d \quad \text{and} \quad E\left[\,\hat{\mathrm{t}}_{\mathrm{d}}^2 \mid n_r\right] = \frac{\mathrm{N}^2}{n_r} S_{yd}^2 + \sum\sum_{j \neq k} \frac{N y_{dj} y_{dk}}{(N-1)},$$

so that the corresponding variance is

$$V\left[N \sum_{S_r} \frac{y_{dk}}{n_r}\right] = N^2 \left[E\left(n_r^{-1}\right) - \frac{1}{N}\right] S_{yd}^2 \qquad (2.1.3)$$

where n_r is a truncated binomial with support on $\{1, \dots, N\}$. More generally, the same fundamental behaviour remains valid if we choose to reject the event $\{n_r < m\}$. In that case the estimator remains unbiased and the variance is still given by expression (2.1.3), except that now the respondent sample size n_r is distributed as a truncated binomial with support on $\{m, \dots, N\}$.

Moreover, our variance computations can be extended to the case where we have this general rejecting Bernoulli design combined with MAR non-response. As in the SRS case, the estimator is now slightly biased so (2.1.3) is no longer valid, but the basic formula (2.1.2) still holds. In this context, conditioning on the nominal (random) sample size n and using the total probability formula we obtain that n_r is now distributed as a compound binomial with

$$\mathrm{P}\{n_r = \mathrm{k}\} = \frac{\mathrm{P}\{n^* = \mathrm{k}\}}{\mathrm{P}\{n \geq \mathrm{m}\}}, k = 1, \dots, N, \;\; \mathrm{P}\{n_r = 0\} = 1 - \frac{\mathrm{P}\{n^* \geq 1\}}{\mathrm{P}\{n \geq \mathrm{m}\}},$$

where $n \sim Bin(N, \pi)$ and $n^* \sim Bin(N, \pi\pi_r)$ results from the combination of Bernoulli selection with probability π and MAR non-response with constant rate π_r .

2.3. 2-phase SRS with MAR non-response

Let us develop now the formulas for the 2-phase stratified SRS with simple non-response at both phases and using the corresponding total estimator $\hat{t}_{d^*} = \sum_{S_r} \frac{y_{dk}}{\alpha_k \beta_{k|a}}$. To make the derivation more amenable, we will proceed by incremental steps of complexity.

We begin drawing our first-phase sample s_a of size n_a from U, which we stratify as $s_a = \cup_{h=1}^{H} s_{ah}$ with each stratum being of size N_h so that $n_a = \sum_{h=1}^{H} N_h$. In each second-phase stratum s_{ah} we draw an SRS s_h of size n_h. At each phase we will assume a simple non-response mechanism homogeneous and independent in each stratum. This may not be a completely realistic non-response model, but it is useful in our context and allows us to cover a variety of designs. To keep the notation simple we will not put the suffix the suffix r everywhere: unless stated otherwise, all sample indexes and counts will in fact represent respondent units.

The estimator we use is the doubly-expanded total estimator calibrated for counts $\hat{t}_{d^*} = \sum_s \frac{y_{dk}}{\alpha_k \beta_{k|a}}$, where $y_{dk} = y_k \cdot \mathrm{I}(k \in D)$ and coefficients $\alpha_k = n_a/\mathrm{N}$, $\beta_{k|a} = n_h/N_h$, $k \epsilon s_{ah}$.

A word of caution is necessary here because our notation can be misleading: without (second-phase) non-response, n_h and N_h are both entirely determined by the first-phase sample S_a, but with the non-response n_h is no longer S_a measurable. Hence, the weights α_k and $\beta_{k|a}$ are random variables, the first being measurable with respect to S_a but the second is not. Also, to correctly handle the case when denominators are zero we always exclude the corresponding events when taking expectations, hence there will be a slight abuse of notation in some expressions. We first compute the following conditional expectation.

$$\begin{aligned} E\left[\sum_S \frac{y_{dk}}{\alpha_k \beta_{k|a}} | s_a\right] &= E\left[\sum_{S_a} \frac{y_{dk} \mathrm{I}_k}{\alpha_k \beta_{k|a}} | s_a\right] \\ &= \sum_h \sum_{S_{ah}} \frac{y_{dk}}{\alpha_k} E\left[\mathrm{I}_k \, \beta_{k|a}^{-1} | s_a\right] \end{aligned}$$

$$= \sum_h \sum_{S_{ah}} \frac{y_{dk} \mathrm{P}\{m_g > 0 | s_a\}}{\alpha_k} \qquad (2.2.1)$$

Taking the expectation again we obtain

$$E\left[\sum_s \frac{y_{dk}}{\alpha_k \beta_k}\right] = E\left[\sum_h \sum_{S_{ah}} \frac{y_{dk} \mathrm{P}\{n_h > 0 | s_a\}}{\alpha_k}\right] \qquad (2.2.2)$$

$$= E\left[\sum_{S_a} \frac{y_{dk}}{\alpha_k} - \sum_h \mathrm{P}\{n_h = 0 | s_a\} \sum_{S_{ah}} \frac{y_{dk}}{\alpha_k}\right]$$

$$= N\bar{y}_d \mathrm{P}\{n_a > 0\} - E\left[\sum_h \mathrm{P}\{n_h = 0 | s_a\} \sum_{S_{ah}} \frac{y_{dk}}{\alpha_k}\right]$$

where the right hand side term comes from our previous computations (2.1.1) for the simple non-response one-phase SRS case. Provided the probabilities $\mathrm{P}\{n_a = 0\}$ and $\mathrm{P}\{n_h = 0 | s_a\}$ are uniformly small, the resulting bias will be negligible. We now compute the conditional expectation of $\hat{t}_{d^*}^2$ given the first-phase sample.

$$E\left[\left(\sum_s \frac{y_{dk}}{\alpha_k \beta_{k|a}}\right)^2 | \, S_a\right] = E\left[\left(\sum_\mathrm{h} \sum_{S_{ah}} \frac{y_{dk}\, \mathrm{I}_k}{\alpha_k \beta_{k|a}}\right)^2 | \, S_a\right]$$

$$= E\left\{\sum_\mathrm{h}\left[\sum_{S_{ah}} \frac{y_{dk}^2\, \mathrm{I}_k}{\alpha_k^2\, \beta_{k|a}^2} + \sum_{\substack{j \neq k \\ S_{ah}}} \frac{y_{dj} y_{dk}\, \mathrm{I}_j \mathrm{I}_k}{\alpha_j \alpha_k \beta_{j|a} \beta_{k|a}}\right] + \sum_{\mathrm{h} \neq \mathrm{l}}\left[\sum_{S_{ah}} \frac{y_{dj}\, \mathrm{I}_j}{\alpha_j \beta_{j|a}}\right]\left[\sum_{S_{al}} \frac{y_{dk}\, \mathrm{I}_k}{\alpha_k \beta_{k|a}}\right] \; | \, S_a\right\}$$

$$= \sum_\mathrm{h}\left[\sum_{S_{ah}} \frac{y_{dk}^2}{\alpha_k^2} E\left(\frac{\mathrm{I}_k}{\beta_{k|a}^2} | s_a\right) + \sum_{\substack{j \neq k \\ S_{ah}}} \frac{y_{dj} y_{dk}}{\alpha_j \alpha_k} E\left(\frac{\mathrm{I}_j \mathrm{I}_k}{\beta_{j|a} \beta_{k|a}} | s_a\right)\right]$$
$$+ \sum_{\mathrm{h} \neq \mathrm{l}}\left[\sum_{S_{ah}} \frac{y_{dj}}{\alpha_j} E\left(\frac{\mathrm{I}_j}{\beta_{j|a}} | s_a\right)\right]\left[\sum_{S_{al}} \frac{y_{dk}}{\alpha_k} E\left(\frac{\mathrm{I}_k}{\beta_{k|a}} | s_a\right)\right]$$

$$= \sum_\mathrm{h}\left[\sum_{S_{ah}} \frac{y_{dk}^2}{\alpha_k^2} N_h E(\mathrm{n_h^{-1}} | s_a) + \sum_{\substack{j \neq k \\ S_{ah}}} \frac{y_{dj} y_{dk}}{\alpha_j \alpha_k} \frac{N_h}{N_h - 1}\left[\mathrm{P}\{n_h > 0 | s_a\} - E(\mathrm{n_h^{-1}} | s_a)\right]\right]$$
$$+ \sum_{\mathrm{h} \neq \mathrm{l}}\left[\sum_{S_{ah}} \frac{y_{dj}}{\alpha_j} \mathrm{P}\{n_h > 0 | s_a\}\right]\left[\sum_{S_{al}} \frac{y_{dk}}{\alpha_k} \mathrm{P}\{n_l > 0 | s_a\}\right]$$

$$= \sum_\mathrm{h} N_h E(\mathrm{n_h^{-1}} | s_a)\left[\sum_{S_{ah}} \frac{y_{dk}^2}{\alpha_k^2} - \frac{N_h}{N_h - 1} \sum_{\substack{j \neq k \\ S_{ah}}} \frac{y_{dj} y_{dk}}{\alpha_j \alpha_k}\right] + \sum_h \frac{N_h \mathrm{P}\{n_h > 0 | s_a\}}{N_h - 1} \sum_{\substack{j \neq k \\ S_{ah}}} \frac{y_{dj} y_{dk}}{\alpha_j \alpha_k}$$
$$+ \left[\sum_\mathrm{h} \sum_{S_{ah}} \frac{y_{dk} \mathrm{P}\{n_h > 0 | s_a\}}{\alpha_k}\right]^2 - \sum_h \left[\sum_{S_{ah}} \frac{y_{dk} \mathrm{P}\{n_h > 0 | s_a\}}{\alpha_k}\right]^2$$

$$= \sum_\mathrm{h} N_h^2 E(\mathrm{n_h^{-1}} | s_a) S_{h\check{y}_d}^2 + \sum_h \frac{N_h \mathrm{P}\{n_h > 0 | s_a\}}{N_h - 1} \sum_{\substack{j \neq k \\ S_{ah}}} \frac{y_{dj} y_{dk}}{\alpha_j \alpha_k} - \sum_h \left[\sum_{S_{ah}} \frac{y_{dk} \mathrm{P}\{n_h > 0 | s_a\}}{\alpha_k}\right]^2$$
$$+ \left[\sum_\mathrm{h} \sum_{S_{ah}} \frac{y_{dk} \mathrm{P}\{n_h > 0 | s_a\}}{\alpha_k}\right]^2$$

$$= \sum_h N_h^2 E(n_h^{-1}|s_a) S_{h\breve{y}_d}^2 - \sum_h N_h \mathrm{P}\{n_h > 0|s_a\} S_{h\breve{y}_d}^2 + \sum_h \mathrm{P}\{n_h > 0|s_a\}\mathrm{P}\{n_h = 0|s_a\} \left[\sum_{s_{ah}} \frac{y_{dk}}{\alpha_k}\right]^2 + \left[\sum_h \sum_{s_{ah}} \frac{y_{dk}\mathrm{P}\{n_h>0|s_a\}}{\alpha_k}\right]^2$$

$$= \sum_h N_h^2 \left[E(n_h^{-1}|s_a) - \frac{\mathrm{P}\{n_h>0|s_a\}}{N_h}\right] S_{h\breve{y}_d}^2 + \sum_h \mathrm{P}\{n_h > 0|s_a\}\mathrm{P}\{n_h = 0|s_a\}\ N_h^2\, \bar{\breve{y}}_{hd}^2 + \left[\sum_h \mathrm{P}\{n_h > 0|s_a\} N_h \bar{\breve{y}}_{hd}\right]^2 \qquad (2.2.3)$$

Where $S_{h\breve{y}_d}^2 = \frac{1}{N_h - 1}\sum_{s_{ah}}\left(\frac{y_{dk}}{\alpha_k} - \bar{\breve{y}}_{hd}\right)^2$, $\bar{\breve{y}}_{hd} = \frac{1}{N_h}\sum_{s_{ah}}\frac{y_{dk}}{\alpha_k}$

Now we can start computing the sampling variance for $\hat{t}_{d^*}$

$$V\left[\sum_s \frac{y_{dk}}{\alpha_k \beta_k}\right] = E\left\{\sum_h N_h^2 \left[E(n_h^{-1}|s_a) - \frac{\mathrm{P}\{n_h>0|s_a\}}{N_h}\right] S_{h\breve{y}_d}^2 + \sum_h \mathrm{P}\{n_h > 0|s_a\}\mathrm{P}\{n_h = 0|s_a\} N_h^2\, \bar{\breve{y}}_{hd}^2\right\} + E\left[\left[\sum_h \sum_{s_{ah}} \frac{y_{dk}\mathrm{P}\{n_h>0|s_a\}}{\alpha_k}\right]^2\right] - E^2\left[\sum_h \sum_{s_{ah}} \frac{y_{dk}\mathrm{P}\{n_h>0|s_a\}}{\alpha_k}\right] \qquad (2.2.4)$$

The last two terms can be seen to form the variance of the conditional expectation $E\left[\sum_s \frac{y_{dk}}{\alpha_k \beta_k}|s_a\right]$. From our previous computations we obtain

$$V\left[E\left[\sum_s \frac{y_{dk}}{\alpha_k \beta_k}|s_a\right]\right] = V\left[\sum_{s_a} \frac{y_{dk}}{\alpha_k} - \sum_h \mathrm{P}\{n_h = 0|s_a\}\sum_{s_{ah}}\frac{y_{dk}}{\alpha_k}\right] \qquad (2.2.5)$$

$$= V\left[\sum_{s_a}\frac{y_{dk}}{\alpha_k}\right] + V\left[\sum_h \mathrm{P}\{n_h = 0|s_a\}\sum_{s_{ah}}\frac{y_{dk}}{\alpha_k}\right] - 2Cov\left\{\sum_{s_a}\frac{y_{dk}}{\alpha_k}, \sum_h \mathrm{P}\{n_h = 0|s_a\}\sum_{s_{ah}}\frac{y_{dk}}{\alpha_k}\right\}$$

$$= N^2\left[E(n_a^{-1}) - \frac{\mathrm{P}\{n_a>0\}}{N}\right] S_{yd}^2 + \mathrm{P}\{n_a > 0\}\mathrm{P}\{n_a = 0\} N^2 \bar{y}_d^2 + V\left[\sum_h \mathrm{P}\{n_h = 0|s_a\}\sum_{s_{ah}}\frac{y_{dk}}{\alpha_k}\right] - 2Cov\left\{\sum_{s_a}\frac{y_{dk}}{\alpha_k}, \sum_h \mathrm{P}\{n_h = 0|s_a\}\sum_{s_{ah}}\frac{y_{dk}}{\alpha_k}\right\}$$

Finally, we obtain the complete expression

$$V\left[\sum_s \frac{y_{dk}}{\alpha_k \beta_k}\right] = N^2\left[E(n_a^{-1}) - \frac{\mathrm{P}\{n_a>0\}}{N}\right] S_{yd}^2 + \mathrm{P}\{n_a > 0\}\mathrm{P}\{n_a = 0\} N^2 \bar{y}_d^2 \qquad (2.2.6)$$

$$+ V\left[\sum_h \mathrm{P}\{n_h = 0|s_a\}\sum_{s_{ah}}\frac{y_{dk}}{\alpha_k}\right] - 2Cov\left\{\sum_{s_a}\frac{y_{dk}}{\alpha_k}, \sum_h \mathrm{P}\{n_h = 0|s_a\}\sum_{s_{ah}}\frac{y_{dk}}{\alpha_k}\right\}$$

$$+ E\left\{\sum_h N_h^2\left[E(n_h^{-1}|s_a) - \frac{\mathrm{P}\{n_h>0|s_a\}}{N_h}\right] S_{h\breve{y}_d}^2 + \sum_h \mathrm{P}\{n_h > 0|s_a\}\mathrm{P}\{n_h = 0|s_a\} N_h^2\, \bar{\breve{y}}_{hd}^2\right\}$$

At this point we can introduce the first phase stratification in our model. It is not hard to see how our previous computations have to be modified in order to accommodate independent first-phase subsamples in a partition $U_1, \dots, U_g, \dots, U_G$ of the initial population. Mutatis mutandis, the resulting variance remains essentially of the same form

$$V\left[\sum_s \frac{y_{dk}}{\alpha_k \beta_k}\right] = \sum_{\mathrm{g}} \left\{ M_g^2 \left[E\left(\mathrm{n}_{\mathrm{ag}}^{-1}\right) - \frac{\mathrm{P}\{n_{ag}>0\}}{M_g} \right] S_{gy_d}^2 + \mathrm{P}\{n_{ag} > 0\}\mathrm{P}\{n_{ag} = 0\} M_g^2 \bar{y}_{dg}^2 \right\} \qquad (2.2.7)$$

$$+ V\left[\sum_h \mathrm{P}\{n_h = 0 | s_a\} \sum_{s_{ah}} \frac{y_{dk}}{\alpha_k}\right] - 2Cov\left\{\sum_{s_a} \frac{y_{dk}}{\alpha_k}, \sum_h \mathrm{P}\{n_h = 0 | s_a\} \sum_{s_{ah}} \frac{y_{dk}}{\alpha_k}\right\}$$

$$+ E\left\{\sum_{\mathrm{h}} N_h^2 \left[E\left(\mathrm{n}_{\mathrm{h}}^{-1} | s_a\right) - \frac{\mathrm{P}\{n_h>0|s_a\}}{N_h}\right] S_{h\check{y}_d}^2 + \sum_h \mathrm{P}\{n_h > 0 | s_a\}\mathrm{P}\{n_h = 0 | s_a\} N_h^2 \, \bar{\check{y}}_{hd}^2 \right\}$$

where $S_{gy_d}^2 = \frac{1}{M_g - 1} \sum_{U_g} \left(y_{dk} - \bar{y}_{dg}\right)^2$, $\bar{y}_{dg} = \frac{1}{M_g} \sum_{U_g} y_{dk}$ and n_{ag} is the number of respondents to the 1st phase sample in stratum U_g.

As we saw previously with the simple one-phase case, there are several nuisance terms that appear when the probabilities $\mathrm{P}\{n_{ag} = 0\}$ and $\mathrm{P}\{n_h = 0 | s_a\}$ are positive. In general, we cannot avoid them since the non-response mechanism makes the corresponding events possible.

However, it appears that even large amounts of non-response could be handled at the first phase, in the sense that (2.2.7) still allows us to compute the resulting variance. But this is not so at the second phase, where the amount of non-response we can deal with corresponds only to the situation when the probabilities $\mathrm{P}\{n_h = 0 | s_a\}$ are negligible. According to our simulations, a simple and effective way to achieve this is to have $E(n_h | s_a) \geq 10$ for all second-phase strata. In most situations this would ensure that coefficients of variation remain within 10% of their nominal level for levels as low as 2%. Even so, second phase non-response will still have an impact on the variance through the conditional expectations $E\left(\mathrm{n}_{\mathrm{h}}^{-1} | s_a\right)$.

Assuming the nuisance terms can safely be ignored, we have the approximation

$$V\left[\sum_s \frac{y_{dk}}{\alpha_k \beta_{k|a}}\right] \cong \sum_{\mathrm{g}} \left\{ M_g^2 \left[E\left(\mathrm{n}_{\mathrm{ag}}^{-1}\right) - \frac{\mathrm{P}\{n_{ag}>0\}}{M_g} \right] S_{gy_d}^2 + \mathrm{P}\{n_{ag} > 0\}\mathrm{P}\{n_{ag} = 0\} M_g^2 \bar{y}_{dg}^2 \right\}$$

$$+ E\left[\sum_{\mathrm{h}} N_h^2 \left[E\left(\mathrm{n}_{\mathrm{h}}^{-1} | s_a\right) - \frac{1}{N_h}\right] S_{h\check{y}_d}^2\right] \qquad (2.2.8)$$

To compute the second term requires knowing how the second phase strata will be formed for every possible s_a. In practice, this is done only for the realized s_a; even if this was available for each first-phase sample, attempting to formalize it would be almost hopeless in most practical cases. As this information is not known, we cannot make further progress unless we make additional simplifying hypotheses. Here it is customary to work conditionally on a realized first-phase sample, assuming it represents reasonably well the expectation— for another approach, see (Rao1973).

These considerations lead to the following expression for the sampling variance of the domain total calibrated for counts and adjusted for MAR non-response at both phases. Now that the computations have been carried out, we can use a more precise notation and indicate explicitly that we work with respondents:

$$V\left[\sum_{s_r} \frac{y_{dk}}{\alpha_k \beta_{k|a}}\right] \cong \sum_{\mathrm{g}} \left\{ M_g^2 \left[E\left(\mathrm{n}_{\mathrm{agr}}^{-1}\right) - \frac{\mathrm{P}\{n_{agr}>0\}}{M_g} \right] S_{gy_d}^2 + \mathrm{P}\{n_{agr} > 0\} \mathrm{P}\{n_{agr} = 0\} M_g^2 \bar{y}_{dg}^2 \right\} + \sum_{\mathrm{h}} N_{hr}^2 \left[E\left(\mathrm{n}_{\mathrm{hr}}^{-1} | s_{ar}\right) - \frac{1}{N_{hr}} \right] S_{hr\check{y}_d}^2 \qquad (2.2.9)$$

As we have mentioned before, the above computations are valid as long as the inclusion indicators are exchangeable. In particular, they hold for the general rejecting Bernoulli schemes with MAR non-response discussed previously. The only elements that need to be adjusted when working under those designs is the distribution of n_{agr} and the conditional distribution of n_{hr} given s_{ar} (see details at the end of section 2.1).

2.4. Variance for Allocation

Once we have obtained an expression we can compute, there is still some work involved in order to make it amenable to the numerical optimization methods that will be used in the allocation problems. The variance (2.2.9) contains the terms $E\left(\mathrm{n}_{\mathrm{agr}}^{-1}\right)$ and $E(\mathrm{n}_{\mathrm{hr}}^{-1}|s_{ar})$ which correspond to the expectation of a reciprocal binomial. These expectations cannot be evaluated directly (there is no close form) and have to be approximated. The usual approximation that consists in taking the inverse of the expectation is not satisfactory for our purposes as it can underestimate the true value by a significant margin when the sampling fraction is small.

Starting from the identity $X^{-1} = (1 + X^{-1})(X + 1)^{-1}$ and replacing the term X^{-1} that appears on the right by a suitable constant, we approximate the initial problem by evaluating a similar expectation that admits a close form in the binomial distribution family. Again, by convention we sum only on the portion of the support that makes the initial expectation meaningful.

$$E[X^{-1}] \cong (1 + \xi^{-1}) E[(X + 1)^{-1}] = (1 + \xi^{-1}) \sum_{k=1}^{N} \frac{N!}{k!(N-k)!} \frac{1}{(k+1)} * \pi^k (1 - \pi)^{N-k}$$

$$= \frac{1+\xi^{-1}}{\pi(N+1)} \sum_{k=1}^{N} \frac{(N+1)!}{(k+1)!(N-k)!} \pi^{k+1} (1-\pi)^{N-k}$$

$$= \frac{1+\xi^{-1}}{\pi(N+1)} \sum_{k=2}^{N+1} \binom{N+1}{k} \pi^{k} (1-\pi)^{N+1-k}$$

$$= \frac{1+\xi^{-1}}{\pi(N+1)} \left[1-(1-\pi)^{N+1} - (N+1)\pi(1-\pi)^{N}\right] \qquad (2.3.1)$$

Numerical simulations show that the choice $\xi = \frac{N\pi}{1.4-0.4\pi}$ gives excellent results when $N \geq 5$ and $N\pi \geq 2$. Dropping the exponential terms yields almost equally satisfactory results when $N\pi \geq 4$:

$$E[X^{-1}] \cong \frac{N\pi+1.4-0.4\pi}{\pi^2 N(N+1)} \qquad (2.3.2)$$

Le tableau ci-contre montre les différences relatives entre la valeur approchée et la vraie valeur en utilisant respectivement (2.3.1), (2.3.2) et une troisième approximation qui correspond à l'inverse de l'espérance.

The following table shows the relative differences between the approximate value and the true value for (2.3.1), (2.3.2) and a third method which corresponds to the inverse of the expectation.

Table 2.3.3

N \ π	0.2	0.4	0.6	0.8
5	Approx1= 25% Approx2= 264% Approx3= 88%	Approx1= - 2.4% Approx2= 27% Approx3= - 5.7%	Approx1= - 4.7% Approx2= - 0.6% Approx3= - 14 %	Approx1= - 1.5% Approx2= - 1.3% Approx3= - 6.8%
10	Approx1= -0.7 % Approx2= 46.5 % Approx3= -2.9 %	Approx1= -4.2% Approx2= -1.2% Approx3= -17.0%	Approx1= -0.6% Approx2= -0.5% Approx3= -8.3%	Approx1= 0.2% Approx2= 0.2% Approx3= - 2.9%
20	Approx1= -4.9% Approx2= 0.9% Approx3= -20.3%	Approx1= 0.21% Approx2= 0.24% Approx3= -8.9 %	Approx1= 0.6% Approx2= 0.6% Approx3= -3.7 %	Approx1= 0.3% Approx2= 0.3% Approx3= - 1.3%
30	Approx1= -1.9% Approx2= -1.1% Approx3= -16.2%	Approx1= 0.9% Approx2= 0.9% Approx3= -5.5%	Approx1= 0.6% Approx2= 0.6% Approx3= -2.4%	Approx1= 0.3% Approx2= 0.3% Approx3= - 1%

Despite the fact that the variance formulae developed in section 2.2 allow us to handle a great variety of sampling strategies, the current implementation in G-SAM does not use the full strength of (2.2.9). In particular, in two-phase designs only the last phase of non response is available and we assume that the nuisance terms related to both non-response phases are negligible.

2.5. Optimal Allocation

Two complementary optimal allocation strategies are used in G-SAM. The so-called cost allocation consists in minimizing the sample cost $\sum_h c_h n_h$ under fixed precision constraints, whereas the power allocation method aims at minimizing the loss $\sum_h X_h^{2q}\, CV^2(\hat{Y}_h)$ under fixed sample size. While the basic recipes have been known for a long time in the survey sampling literature, the current implementation involves some new elements that we hope will be useful for many users.

In our context we allow some additional flexibility by using a slightly different formulation, so our cost allocation will use the objective function $\sum_h \sum_j X_{jh} n_h$ and our power allocation will use the loss function $\sum_h \sum_j X_{hj}^{2q}\, Var(\hat{Y}_{jh}|S_a)^p$. Here the quantities X_{jh} represent the relative importance of variable j in stratum h and the variances correspond to one of the sampling strategies covered by formula (2.2.9).

For the estimation strategy consisting of stratified SRS and the Horvitz-Thompson estimator, the cost allocation is a standard case of linear minimization with the simplex method, whereas the power allocation admits a close form solution that can be found using Lagrange multipliers. Even though we will be working with similar minimization problems, since the designs are more complex and the constraints more varied, numerical optimization methods for non-linear problems will be required in order to solve the allocation problems that arise in our context.

In all our allocation problems we consider a variety of constraints that consist of two basic types: precision constraints (CV's) and size constraints. Precision constraints aim at satisfying, for the variables of interest Y_j, target coefficients of variation that can be defined at various levels: global, strata and domain. The size constraints allow the user to specify bounds regarding the desired number of units (or cost) globally and for particular strata.

Many quantities are common to the various methods. Among these, the population variance takes different forms; it seems appropriate to recall them here:

$S_{gy_j}^2 = \frac{1}{M_g - 1}\sum_{U_g}\left(y_{jk} - \bar{y}_{jg}\right)^2, \bar{y}_{jg} = \frac{1}{M_g}\sum_{U_g} y_{jk}$, where y_{jk} is the value of the characteristic of interest y_j for population unit k.

$S_{h\breve{y}_j}^2 = \frac{1}{N_h - 1}\sum_{S_{ah}}\left(\frac{y_{jk}}{\alpha_k} - \bar{\breve{y}}_{jh}\right)^2,\ \bar{\breve{y}}_{jh} = \frac{1}{N_h}\sum_{S_{ah}}\frac{y_{jk}}{\alpha_k},\ \alpha_k = \frac{n_{ag}}{M_g}, k \in U_g$ and y_{jk} is defined as before.

$S_{gy_{jh}}^2 = \frac{1}{M_g - 1}\sum_{U_g}\left(y_{jhk} - \bar{y}_{jhg}\right)^2$, $\bar{y}_{jhg} = \frac{1}{M_g}\sum_{U_g} y_{jhk}$, where $y_{jhk} = y_{jk} \cdot \mathrm{I}(k \in S_{ah})$.

$S_{gy_{jd}}^2 = \frac{1}{M_g - 1}\sum_{U_g}\left(y_{jdk} - \bar{y}_{jdg}\right)^2$, $\bar{y}_{jdg} = \frac{1}{M_g}\sum_{U_g} y_{jdk}$, where $y_{jdk} = y_{jk} \cdot \mathrm{I}(k \in d)$.

$S_{h\breve{y}_{jd}}^2 = \frac{1}{N_h - 1}\sum_{S_{ah}}\left(\frac{y_{jdk}}{\alpha_k} - \bar{\breve{y}}_{jdh}\right)^2,\ \bar{\breve{y}}_{jdh} = \frac{1}{N_h}\sum_{S_{ah}}\frac{y_{jdk}}{\alpha_k},\ \alpha_k = \frac{n_{ag}}{M_g}, k \in U_g$ and y_{jdk} is defined as before.

Population totals and CV's are also common quantities to all methods:

$Y_j = N\bar{y}_j = \sum_{\mathrm{U}} y_{jk}$, $Y_{jh} = N\bar{y}_{jh} = \sum_{\mathrm{U}} y_{jhk}$, $Y_{jd} = N\bar{y}_{jd} = \sum_{\mathrm{U}} y_{jdk}$, $CV_j^2 = \frac{1}{N-1}\sum_{\mathrm{U}}\left(y_{jk} - \bar{y}_j\right)^2 \ / \ Y_j^2$

$CV_{jh}^2 = \frac{1}{N-1}\sum_{\mathrm{U}}\left(y_{jhk} - \bar{y}_{jh}\right)^2 \ / \ Y_{jh}^2$, $CV_{jd}^2 = \frac{1}{N-1}\sum_{\mathrm{U}}\left(y_{jdk} - \bar{y}_{jd}\right)^2 \ / \ Y_{jd}^2$

2.5.1. Cost allocation for two-phase stratified SRS

When using the cost allocation method we are solving for stratum sample sizes n_h that result in the minimum weighted total sample size under a given set of constraints:

Minimize $\sum_h \sum_j X_{jh} n_h$

Subject to

$$\sum_{\mathrm{g}}\left\{M_g^2\left[\frac{1}{m_g} - \frac{1}{M_g}\right]S_{gy_j}^2\right\} + \sum_{\mathrm{h}} N_h^2\left[\frac{(n_h - 0.4)\rho_h + 1.4}{\rho_h^2 n_h(n_h + 1)} - \frac{1}{N_h}\right]S_{h\breve{y}_j}^2 \leq CV_j^2\, Y_j^2\ ,\ j = 1,2,\dots,J$$

$$\sum_{\mathrm{g}}\left\{M_g^2\left[\frac{1}{m_g} - \frac{1}{M_g}\right]S_{gy_{jh}}^2\right\} + N_h^2\left[\frac{(n_h - 0.4)\rho_h + 1.4}{\rho_h^2 n_h(n_h + 1)} - \frac{1}{N_h}\right]S_{h\breve{y}_j}^2 \leq CV_{jh}^2\, Y_{jh}^2\ , h = 1,2,\dots,H$$

$$\sum_{\mathrm{g}}\left\{M_g^2\left[\frac{1}{m_g} - \frac{1}{M_g}\right]S_{gy_{jd}}^2\right\} + \sum_{\mathrm{h}} N_h^2\left[\frac{(n_h - 0.4)\rho_h + 1.4}{\rho_h^2 n_h(n_h + 1)} - \frac{1}{N_h}\right]S_{\mathrm{h}\breve{y}_{jd}}^2 \leq CV_{jd}^2\, Y_{jd}^2\ , d \in \{d_k\}$$

$$c_h n_h \leq MaxCost_h\ ,\ h = 1,2,\dots,H$$

$$\sum_h c_h n_h \leq MaxCost$$

$$n_h \leq MaxSize_h\ ,\ \ h = 1,2, \dots , H$$
$$\textstyle\sum_h n_h \leq MaxSize$$
$$Prob(a_h \leq n_{hr}) \geq r_h\ ,\ \ h = 1,2, \dots , H$$

Assuming full response at the 1st phase, the left term of the precision constraints correspond to an approximation for the sampling variance of the estimator for the domain total calibrated on counts, where each constraint controls the CV of the variable Y_j at a given level. These levels correspond respectively to population, stratum h and domain d. A word of caution is necessary at this point since for a general 2-phase design, the strata that form the second phase stratification are usually defined only after we observe the first phase sample. We can specify precisions at the (2nd phase) stratum h level as long as the characteristics that define this group exist independently of the first phase sample.

To illustrate this concept let us consider a simple example. Suppose the first phase strata correspond to *region* and the second-phase strata to *region*revenue,* where *revenue* is a categorical variable taking on the values *high*/*average*/*low*. If *high=more than 10^5* , *low=less than 10^4* and *average= else* , this case falls in the above category where we can compute the variance for this domain –actually, the precision constraint is not for the stratum itself but rather for the domain (in the population) which restricted to S_a forms the stratum. However, if *high=top 20%, low=lower 20%, average=else,* this now depends on S_a and therefore cannot be handled.

If the 1st-phase variance component for each level is denoted V_{0j}, V_{0jh}, V_{0jd} respectively, the problem can be written as

$$\text{Minimize } \textstyle\sum_h \sum_j X_{jh} n_h$$

Subject to

$$\textstyle\sum_{\mathrm{h}} N_h^2 \left[\frac{(n_h - 0.4)\rho_h + 1.4}{\rho_h^2 n_h (n_h + 1)} - \frac{1}{N_h}\right] S^2_{h\check{y}_j} \leq CV_j^2\, Y_j^2 - V_{0j}\ ,\ \ j = 1,2, \dots , J$$

$$N_h^2 \left[\frac{(n_h - 0.4)\rho_h + 1.4}{\rho_h^2 n_h (n_h + 1)} - \frac{1}{N_h}\right] S^2_{h\check{y}_j} \leq CV_{jh}^2\, Y_{jh}^2 - V_{0jh}\ ,\ \ h = 1,2, \dots , H$$

$$\textstyle\sum_{\mathrm{h}} N_h^2 \left[\frac{(n_h - 0.4)\rho_h + 1.4}{\rho_h^2 n_h (n_h + 1)} - \frac{1}{N_h}\right] S^2_{\mathrm{h}\check{y}_{jd}} \leq\ CV_{jd}^2\, Y_{jd}^2 - V_{0jd}\ ,\ \ d \in \{d_k\}$$

$$c_h n_h \leq MaxCost_h\ ,\ \ h = 1,2, \dots , H$$
$$n_h \leq MaxSize_h\ ,\ \ h = 1,2, \dots , H$$
$$\textstyle\sum_h c_h n_h \leq MaxCost$$
$$\textstyle\sum_h n_h \leq MaxSize$$
$$Prob(a_h \leq n_{hr}) \geq r_h,\ \ h = 1,2, \dots , H$$

In most cases it would be advisable to drop the constraints on total sample cost and total sample size, and let the objective function determine the size required to achieve the desired precision at various levels. The last constraint ensures that the probability of getting at least a_h respondents in each stratum is sufficiently high, where the number of respondents n_{hr} is distributed as a binomial random variable with parameters n_h and ρ_h. As this probability is a monotone increasing function of n_h, the constraint is implemented as $n_h \ge \gamma(\, a_h, \rho_h, r_h)$ where the right hand side corresponds to the minimum value for which the condition holds–this inverse quantile can be determined numerically using standard SAS tools.

This procedure can also be used to allocate a single phase stratified SRS. To do this we only need to set all the 1st-phase variance components V_{0j}, V_{0jh}, V_{0jd} to zero and the 1st-phase selection probabilities $\alpha_k \equiv 1$, which is another way of saying that the first phase sample is in fact a census.

2.5.2. Cost allocation for two-phase stratified BER

When using the cost allocation method we are solving for selection probabilities π_h that result in the minimum expected weighted total sample size under a given set of constraints:

$$\text{Minimize } \sum_h \sum_j X_{hj} N_h \pi_h$$

Subject to

$$\sum_g \left\{ M_g^2 \left[\frac{1}{M_g \pi_{ag}} - \frac{1}{M_g}\right] S^2_{gy_j} \right\} + \sum_h N_h^2 \left[\frac{(N_h-0.4)\pi_h\rho_h+1.4}{\pi_h^2\rho_h^2 N_h(N_h+1)} - \frac{1}{N_h}\right] S^2_{h\check{y}_j} \le CV_j^2\, Y_j^2\,,\ j = 1,2,\dots,J$$

$$\sum_g \left\{ M_g^2 \left[\frac{1}{M_g \pi_{ag}} - \frac{1}{M_g}\right] S^2_{gy_{jh}} \right\} + N_h^2 \left[\frac{(N_h-0.4)\pi_h\rho_h+1.4}{\pi_h^2\rho_h^2 N_h(N_h+1)} - \frac{1}{N_h}\right] S^2_{h\check{y}_j} \le CV_{jh}^2\, Y_{jh}^2\,, h = 1,2,\dots,H$$

$$\sum_g \left\{ M_g^2 \left[\frac{1}{M_g \pi_{ag}} - \frac{1}{M_g}\right] S^2_{gy_{jd}} \right\} + \sum_h N_h^2 \left[\frac{(N_h-0.4)\pi_h\rho_h+1.4}{\pi_h^2\rho_h^2 N_h(N_h+1)} - \frac{1}{N_h}\right] S^2_{h\check{y}_{jd}} \le CV_{jd}^2\, Y_{jd}^2\,, d \in \{d_k\}$$

$$Prob(c_h n_h \le MaxCost_h) \ge r_h$$
$$Prob(n_h \le MaxSize_h) \ge r_h$$
$$Prob(a_h \le n_{hr}) \ge r_h$$
$$Prob(\textstyle\sum_h c_h n_h \le MaxCost) \ge r$$
$$Prob(\textstyle\sum_h c_h n_h \le MaxCost) \ge r$$

With the 1st-phase variance components denoted V_{0j}, V_{0jh}, V_{0jd}, the problem becomes

$$\text{Minimize } \sum_h \sum_j X_{hj} N_h \pi_h$$

Subject to

$$\sum_{\mathrm{h}} N_h^2 \left[\frac{(N_h-0.4)\pi_h\rho_h+1.4}{\pi_h^2\rho_h^2 N_h(N_h+1)} - \frac{1}{N_h}\right] S^2_{h\check{y}_j} \leq CV_j^2\, Y_j^2 - V_{0j}\ ,\ j = 1,2,\dots,J$$

$$N_h^2 \left[\frac{(N_h-0.4)\pi_h\rho_h+1.4}{\pi_h^2\rho_h^2 N_h(N_h+1)} - \frac{1}{N_h}\right] S^2_{h\check{y}_j} \leq CV_{jh}^2\, Y_{jh}^2 - V_{0jh}\ , h = 1,2,\dots,H$$

$$\sum_{\mathrm{h}} N_h^2 \left[\frac{(N_h-0.4)\pi_h\rho_h+1.4}{\pi_h^2\rho_h^2 N_h(N_h+1)} - \frac{1}{N_h}\right] S^2_{\mathrm{h}\check{y}_{jd}} \leq\ CV_{jd}^2\, Y_{jd}^2 - V_{0jd}\ , d \in \{d_k\}$$

$$Prob(c_h n_h \leq MaxCost_h) \geq r_h\ , h = 1,2,\dots,H$$
$$Prob(n_h \leq MaxSize_h) \geq r_h\ , h = 1,2,\dots,H$$
$$Prob(a_h \leq n_{hr}) \geq r_h\ , h = 1,2,\dots,H$$
$$Prob(\textstyle\sum_h c_h n_h \leq MaxCost) \geq r$$
$$Prob(\textstyle\sum_h n_h \leq MaxSize) \geq r$$

The constraints remain basically of the same type as for the SRS case, but since the sample size is random, the resource constraints are now expressed in terms of probabilities–this choice allowing a better control than the corresponding approach with expectations. Among the probability constraints, the ones controlling the stratum size can be handled relatively easily since the probability term is a monotone function of π_h, and we can use a simple bisection method to determine the value that achieves equality. The two global size constraints are solved by using the normal approximation for the distribution of the sum, so they become

$$MaxCost - \textstyle\sum_h N_h \pi_h c_h \geq\ q_r \sqrt{\textstyle\sum_h\ c_h^2 N_h \pi_h (1-\pi_h)}$$

$$MaxSize - \textstyle\sum_h N_h \pi_h \geq\ q_r \sqrt{\textstyle\sum_h N_h \pi_h (1-\pi_h)}$$

where q_r is the quantile of order r of the standard normal.

All remarks formulated for the cost allocation with SRS apply here as well, except that now the number of respondents n_{hr} is a binomial random variable with parameters N_h and $\pi_h\rho_h$. As before, the method can be used to allocate a single phase stratified BER by setting the 1st-phase variance components V_{0j}, V_{0jh}, V_{0jd} to zero and the 1st-phase selection probabilities α_k to 1.

2.5.3. Power Allocation for two-phase Stratified SRS

When using the power allocation method we are solving for stratum sample sizes n_h that result in the minimum total weighted conditional loss:

$$\text{Minimize } \sum_h \sum_j X_{hj}^{2q} \left[N_h^2 \left(\frac{(n_h-0.4)\rho_h+1.4}{\rho_h^2 n_h(n_h+1)} - \frac{1}{N_h}\right) S_{h\check{y}_j}^2\right]^p$$

Subject to

$$\sum_{\mathrm{h}} N_h^2 \left[\frac{n_h\rho_h+1.4-0.4\rho_h}{\rho_h^2 n_h(n_h+1)} - \frac{1}{N_h}\right] S_{h\check{y}_j}^2 \leq CV_j^2\, Y_j^2 - V_{0j}\ ,\ j = 1,2,\dots,J$$

$$N_h^2 \left[\frac{n_h\rho_h+1.4-0.4\rho_h}{\rho_h^2 n_h(n_h+1)} - \frac{1}{N_h}\right] S_{h\check{y}_j}^2 \leq CV_{jh}^2\, Y_{jh}^2 - V_{0jh}\ ,\ h = 1,2,\dots,H$$

$$\sum_{\mathrm{h}} N_h^2 \left[\frac{n_h\rho_h+1.4-0.4\rho_h}{\rho_h^2 n_h(n_h+1)} - \frac{1}{N_h}\right] S_{\mathrm{h}\check{y}_{jd}}^2 \leq\ CV_{jd}^2\, Y_{jd}^2 - V_{0jd}\ ,\ d \in \{d_k\}$$

$$c_h n_h \leq MaxCost_h\ ,\ \ h = 1,2,\dots,H$$
$$n_h \leq MaxSize_h\ ,\ \ h = 1,2,\dots,H$$
$$\sum_h c_h n_h \leq MaxCost$$
$$\sum_h n_h \leq MaxSize$$
$$Prob(a_h \leq n_{hr}) \geq r_h\ ,\ \ h = 1,2,\dots,H$$

All quantities are exactly the same as the corresponding ones developed in 3.1 for the case of cost allocation under 2-phase stratified SRS. As always, the method can be used to allocate a single phase stratified SRS by setting the 1st-phase variance components V_{0j}, V_{0jh}, V_{0jd} to zero and the 1st-phase selection probabilities α_k to 1.

2.5.4. Power Allocation for two-phase Stratified BER

When using the power allocation method we are solving for stratum sample probabilities π_h that result in the minimum total weighted conditional loss:

$$\text{Minimize } \sum_h \sum_j X_{hj}^{2q} \left[N_h^2 \left(\frac{(N_h - 0.4)\pi_h \rho_h + 1.4}{\pi_h^2 \rho_h^2 N_h (N_h + 1)} - \frac{1}{N_h} \right) S_{h\check{y}_j}^2 \right]^p$$

Subject to

$$\sum_{\mathrm{h}} N_h^2 \left[\frac{(N_h - 0.4)\pi_h \rho_h + 1.4}{\pi_h^2 \rho_h^2 N_h (N_h + 1)} - \frac{1}{N_h} \right] S_{h\check{y}_j}^2 \leq CV_j^2 \, Y_j^2 - V_{0j} \ , \ j = 1,2, \ldots , J$$

$$N_h^2 \left[\frac{(N_h - 0.4)\pi_h \rho_h + 1.4}{\pi_h^2 \rho_h^2 N_h (N_h + 1)} - \frac{1}{N_h} \right] S_{h\check{y}_j}^2 \leq CV_{jh}^2 \, Y_{jh}^2 - V_{0jh} \ , \ h = 1,2, \ldots , H$$

$$\sum_{\mathrm{h}} N_h^2 \left[\frac{(N_h - 0.4)\pi_h \rho_h + 1.4}{\pi_h^2 \rho_h^2 N_h (N_h + 1)} - \frac{1}{N_h} \right] S_{\mathrm{h}\check{y}_{jd}}^2 \leq \ CV_{jd}^2 \, Y_{jd}^2 - V_{0jd} \ , \ d \in \{d_k\}$$

$$Prob(c_h n_h \leq MaxCost_h) \geq r_h \ , \ h = 1,2, \ldots , H$$
$$Prob(n_h \leq MaxSize_h) \geq r_h \ , \ h = 1,2, \ldots , H$$
$$Prob(a_h \leq n_{hr}) \geq r_h \ , \ h = 1,2, \ldots , H$$
$$MaxCost - \sum_h N_h \pi_h c_h \geq \ q_r \sqrt{\sum_h \ c_h^2 N_h \pi_h (1 - \pi_h)}$$
$$MaxSize - \sum_h N_h \pi_h \geq \ q_r \sqrt{\sum_h N_h \pi_h (1 - \pi_h)}$$

All quantities are exactly the same as the corresponding ones developed in 3.2 for the case of cost allocation under 2-phase stratified BER. Again, the method can be used to allocate a single phase stratified BER by setting the 1st-phase variance components V_{0j}, V_{0jh}, V_{0jd} to zero and the 1st-phase selection probabilities α_k to 1.

3. Selection and Coordination

The Selection and Coordination module built for G-SAM version 1.0 offers two basic selection schemes: stratified Simple Random Sampling (SRS) and stratified Bernoulli sampling. Both schemes are implemented using the so-called sequential method based on uniform random numbers and selection windows (Ohlsson, 1995). This implementation allows the coordination of samples from different surveys by selecting appropriate starting points on the frame. It can also handle sample rotation between successive occurrences of a repeated survey.

3.1. Sequential SRS and Bernoulli

Given a population U partitioned into strata, this method will independently select a subsample in each stratum according to SRS or Bernoulli sampling. We begin by assuming there is a *uniform random number* (URN) X_k associated to every unit u_k in U, where $(X_k)_{k=1}^{N}$ form an independent sample from the uniform (0, 1) distribution– any continuous distribution could have been used but the uniform is particularly convenient. In the survey literature these random numbers are often called *permanent random numbers* (PRN), since for a given survey frame once they are drawn they remain associated with the corresponding units for all samples that will be selected from that frame.

For SRS, the essential feature of the sample (X_k) is that distribution-wise, it induces equally likely permutations among the populations units u_k . The simplest way of using the random numbers X_k to draw an SRS of size n_h in stratum U_h, is to select the units u_k corresponding to the n_h smallest X_k in the stratum. Similarly, the n_h largest X_k in the stratum will also select an SRS. In fact, as long as the choice is independent of (X_k) any subset of n_h order statistics will select an SRS of that size. A particular case that is widely used in practice is to select n_h values $X_{(k)}$ immediately to the right of a fixed value z_h called the start. When there are less than n_h units determined in this fashion, we supplement the subset with the necessary $X_{(k)}$ immediately to the right of 0–in other words, we wrap around the ends of the stratum. Of course, mutatis mutandis we can also select the units to the left of the start.

To select according to a Bernoulli design with probability π_h in stratum U_h, we can select the units u_k in U_h corresponding to the X_k satisfying the condition $z_h \leq X_k \leq z_h + \pi_h$ (wrapping around the boundaries of the unit interval to achieve the desired length if $z_h + \pi_h > 1$). As the random variables X_k are independent and the above condition holds with probability π_h, this selection scheme does satisfy the required sampling plan. Similarly as for the SRS case, we can also select to the left of the starting point, wrapping around when $z_h - \pi_h < 0$.

3.2. Rotation

Rotation of samples is a technique used for controlling the overlap between consecutive samples in a repeated survey. It consists in randomly selecting a portion of the previous sample and replacing it with units drawn from the rest of the population. The sequential selection mechanism provides us with simple recipes for performing the rotation of samples. Let $X_{(1)}, \dots, X_{(H)}$ be the ordered URN's in a given stratum of size N and suppose that a previous SRS had been selected using $X_{(1)}, \dots, X_{(n)}$. If we apply a shift to the initial ranks and use $X_{(1+k)}, \dots, X_{(n+k)}$ to perform our current selection, then both samples have the same design and the overlap is exactly k/n. Another possibility is to apply a shift to the start, so that if z_0 was used for selecting the previous sample and assuming selection to the right, if our current selection uses $z_0 + r \cdot n/N$ then the samples will be separated by $r \cdot n$ units on average and an expected rotation rate r will be achieved. This is for rotating under an SRS design, but it can easily be adapted to rotate Bernoulli samples by simply moving the initial selection window $r \cdot \pi$ units in the direction of the initial selection.

The method based on shifting ranks allows a much more precise control of the overlap, but cannot be adapted to Bernoulli sampling without restricting its scope considerably. Furthermore, even in the SRS context the rotation rate will seldom be exact due to changes on the frame between the occurrences of the survey. For these reasons the second method was chosen to be implemented in G-SAM– this is also the standard approach used in many other statistical agencies.

Using the starts of the previous survey sample and the current sampling fractions, G-SAM determines the starts that yield the rotation rates specified by the user and selects the new sample. Although it is possible to rotate samples from different designs, the above process assumes that the whole design remains the same. It also assumes the frame and stratification remain the same. Of course this is rarely the case in practice, but as long as these changes are not too pronounced this should not have a significant impact on the results. Besides, even in an ideal situation the specified rotation rate is only satisfied on average.

In general, sample coordination will achieve more consistent results when compared to the coordination of different surveys. This is simply because in a repeated survey the sampling design–including the stratification–remains fairly stable.

3.3. Coordination

Coordination of samples aims at controlling the overlap between multiple samples selected from a given frame. Although no general method for coordinating samples from arbitrary designs exists, the methods based on URN's can accommodate many common sampling plans. If we use independent selections as a reference point, *positive coordination* corresponds to the case where we have more overlap whereas *negative coordination* corresponds to less; in the survey literature these terms are often synonyms for maximum and minimum overlap.

Provided there is enough space on the frame, to negatively coordinate two samples using URN's is quite simple. As we did previously for rotation, the idea is to choose sets of starts which are sufficiently far apart, regardless of the designs. For example, if survey A uses 0 as a start in all strata and selects to the right, survey B could select in the same direction and use as a global start the maximum sampling fraction in survey A. If both surveys use Bernoulli sampling this strategy would ensure no overlap as the selection windows would be disjoint; however, if one of them is an SRS the extent of the overlap will depend on the sampling fractions and the stratifications involved.

Following the same logic, positive coordination of two samples by means of URN's is achieved by selecting in the same direction and choosing sets of starts which are sufficiently close, the extreme case being when we take identical starts. If the stratifications are the same this would select identical samples, but even when they are different this process will generally achieve some degree of positive coordination.

Regarding coordination and rotation, typically these concepts don't mix well. The reason being that coordination will generally be destroyed after a few cycles of rotation. One notable exception is the case when a group of coordinated surveys is shifted by the same amount and in the same direction. In this case each survey is rotated in order to preserve coordination, but rotation rates fluctuate according to the extent of the shift compared to the sampling fractions involved.

A final word of caution is necessary at this point to clarify a number of aspects for the proper use of the coordination tools available in G-SAM. Strictly speaking, the current version is not capable of determining how to coordinate samples: except in the simple case of rotation, it cannot find the starts that would ensure coordination within a group of surveys. Currently there are available coordination methods that would perform automated coordination–notably Cotton&Hesse and MicroStrate–but they are based on different principles and are not compatible with the PRN framework.

Instead, the current version of G-SAM relies on the user to provide the starts that would ensure proper coordination of the current sample with other surveys. However, this goal cannot be satisfactorily achieved unless a certain level of cohesion exists among the surveys for which the coordination is planned–a minimal requirement being that they all use the same set of random numbers to draw their samples. The targeted surveys should also work with stratifications that are compatible and use sampling fractions that facilitate the sharing of the common population units.

For the best results, coordination should be planned right at the beginning of the design stage of a new survey.

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